

Engineering Decision Report

NovaloT FleetGuard — Cold-Chain Container Monitoring Device

Chetan Angadi | NovaloT Embedded Systems Assignment

This report explains the engineering reasoning behind the hardware and architecture selections made for the FleetGuard container monitoring device, the trade-offs considered, what would change for a production deployment, a structured risk analysis, and planned future improvements. It should be read alongside docs/Hardware_Design.md, docs/System_Architecture.md, and docs/Firmware_Architecture.md, which contain the full technical detail this report summarizes and justifies.

1. Why This Hardware Was Selected

1.1 Processing Unit — ESP32-S3-WROOM-1

The ESP32-S3 was chosen over alternatives such as the STM32 family, RP2040, or a plain Arduino board because it uniquely satisfies three requirements at once: sufficient compute headroom for local classification logic running under FreeRTOS, integrated Wi-Fi/BLE for the future connectivity requirement without adding a second radio module, and a mature, well-documented ecosystem that reduces development risk for a first-of-its-kind MVP. STM32 parts offer excellent low-power and real-time characteristics but would require a separate wireless module, adding board area, cost, and integration risk that isn't justified when the workload (periodic sampling, threshold checks, local logging) is not real-time-critical in the hard sense. The RP2040 lacks integrated wireless entirely. The dual-core architecture also allows sensor/alert servicing to be separated from future networking tasks on FreeRTOS without them contending for the same core.

1.2 Environmental Sensor — Sensirion SHT41-AD1B

Temperature and humidity are the two mandatory monitored parameters. The SHT41 was selected for its factory calibration (removing the need for field calibration across thousands of deployed units), its integrated heater (which drives off condensation — a real risk in a refrigerated, humid environment), and its accuracy specification, which is appropriate for perishable-goods compliance reporting. A cheaper DHT22-class sensor was considered and rejected: its accuracy and long-term drift characteristics are not suitable for a device whose readings may need to stand up to a customer dispute about spoiled cargo.

1.3 Optional Sensors

Door status (reed switch), battery voltage (resistive divider + ADC), ambient light (BH1750), and vibration (LIS3DH) were added because each provides low-cost context that materially improves the operator's ability to diagnose why a temperature excursion happened — a door left open explains a temperature spike in a way a bare temperature graph cannot. All four use well-supported, inexpensive, digital interfaces (GPIO/I2C/ADC) that don't meaningfully add to BOM cost or firmware complexity at fleet scale.

1.4 Power Subsystem

The device is assumed to run from 12V/24V vehicle power. Reverse polarity protection (P-channel MOSFET), a resettable PTC fuse, and a TVS diode were included specifically because vehicle electrical systems are electrically noisy and prone to install-time wiring mistakes at scale — for a fleet of thousands of units installed by many different technicians, protecting against a reversed connector is cheaper than the warranty and support cost of not doing so.

1.5 Local Storage

A hybrid of ESP32 NVS (configuration), external SPI flash (event buffer), and an industrial-grade MicroSD card (long-term history) was chosen over a single storage medium to balance write endurance against capacity: the SPI flash buffer absorbs frequent small writes without wearing out the SD card's limited write cycles, while the SD card provides the bulk capacity needed for weeks of unattended operation between depot visits.

2. Why This Architecture Was Selected

2.1 Layered System Architecture

The system is split into six layers (Sensing, Data Acquisition, Decision Engine, Event Logging, Alerting, Future Connectivity) so that each layer has a single, testable responsibility and can evolve independently. In particular, isolating sensor access behind a Data Acquisition layer means the Decision Engine, Event Logger, and Alert Manager never call hardware APIs directly — this is what allowed the entire pipeline to be demonstrated as a software simulation (Task 4, Option B) without any physical board, and it is what will let the same logic run unmodified against real I2C/GPIO/ADC drivers in production firmware.

2.2 Worst-Case-Wins Classification

Rather than a single combined health score, each monitored parameter is classified independently against a two-band threshold (Normal / Warning / Critical), and the overall device condition is the worst individual classification. This was a deliberate, conservative choice: an average or weighted score could let one badly out-of-range parameter be masked by others being fine, which is unacceptable for a safety/compliance-relevant cold-chain product. The two-band structure (rather than a single cutoff) exists to avoid alert chattering right at a boundary and to give operators lead time between Warning and Critical.

2.3 Device State Machine Separate From Condition

Device operating state (BOOT / RUNNING / DEGRADED / FAULT / CONFIG) is modeled separately from environmental condition (Normal / Warning / Critical / Fault). This distinction matters operationally: "the sensor stopped responding" and "the cargo is too warm" require different responses from a technician versus an operations manager, and conflating them in one status field would make the device harder to diagnose in the field.

2.4 Crash-Safe Event Logging

Every event write is flushed and fsync'd before being considered committed, so an abrupt power loss (a realistic event on vehicle power) can lose at most the one in-flight write, never previously committed history. This directly addresses the Reliability non-functional requirement and was validated in the prototype by a scripted power-loss/restart test.

3. Trade-offs Considered

- **Option A (physical hardware) vs Option B (simulation):** Option B was chosen because no physical ESP32-S3 board, sensors, or lab equipment were available in this environment within the assignment's time window. The trade-off is that timing, real bus behaviour, and physical failure modes are not hardware-validated — mitigated by keeping the simulation's module boundaries and logic identical to the intended production firmware.
- **Single MCU edge device vs MCU + gateway:** A self-contained device with local alerting was chosen because it works even when a container is isolated from other units, since local alerting must function with zero connectivity.

- **No local display vs adding a display:** LED + buzzer + serial diagnostics were chosen over a display, trading richer on-device information for lower cost, lower power draw, and less exposure to a harsh environment across a fleet of thousands.
- **Hybrid flash+SD storage vs SD-only:** The SPI flash buffer trades a small amount of extra BOM cost and firmware complexity for meaningfully better long-term reliability at fleet scale versus writing every event directly to the SD card.
- **Threshold-based classification vs ML-based anomaly detection:** A deterministic, configurable threshold model was chosen because it is auditable and explainable to a non-technical operator, and doesn't require training data that doesn't exist yet for a first-of-its-kind deployment.

4. What Would Change For Production Deployment

- **Firmware:** port the simulation's module logic to ESP-IDF/FreeRTOS C, replacing simulated sensor classes with real I2C/GPIO/ADC drivers behind the same interface boundaries.
- **Hardware validation:** full schematic capture and PCB layout, plus EMI/EMC and automotive electrical compliance testing given the 12V/24V vehicle power environment.
- **Security:** secure boot and firmware signing, per-device identity/credentials provisioned at manufacturing time, and TLS for any future cloud communication.
- **OTA updates:** a staged OTA rollout mechanism (canary devices first, automatic rollback on boot failure) is essential once thousands of units are deployed in the field.
- **Manufacturing test:** an automated end-of-line test procedure (sensor presence/communication check, power-on self-test, provisioning) to catch defective units before shipment.
- **Enclosure & certification:** an IP-rated enclosure suitable for a refrigerated, humid, vibration-prone environment, plus applicable regulatory certification.
- **Cloud integration:** implement the MQTT/cloud ingest path the architecture already reserves space for, including device provisioning and fleet-scale device management.

5. Risks and Limitations

- No hardware-in-the-loop validation: real sensor failure modes may behave differently than the simulated fault injection used here.
- Single point of environmental sensing: one SHT41 per container means a physical sensor failure removes the primary monitored signal for that unit until repaired.
- Threshold strategy is generic: default thresholds (2-8°C / 30-70% RH) target chilled goods; other cargo types need their own validated threshold sets before deployment.
- Security is designed-for, not implemented: the current prototype has no authentication on configuration changes.
- Vehicle power quality is assumed, not measured, against an actual target vehicle fleet.

6. Risk Analysis

The table below restates the risks above as a structured Risk / Impact / Mitigation analysis, the format commonly expected in engineering documentation, so each risk is tied to a concrete follow-up action rather than left as a caveat.

Risk	Impact	Mitigation
No hardware-in-the-loop validation (Task 4 delivered as Option B software simulation)	Real sensor bus/timing/failure behaviour may differ from the simulated model, potentially masking issues before field deployment	Port simulation logic 1:1 to ESP-IDF, run hardware-in-the-loop testing on pilot units before mass production
Single point of environmental sensing (one SHT41 per container)	A failed sensor removes the primary monitored signal for that container until physically repaired	Last-known-good fallback with staleness flagging is already implemented; consider a redundant secondary sensor or scheduled maintenance checks for high-value routes
Generic default thresholds not validated per cargo type	Incorrect thresholds could cause false alerts, or worse, miss a genuine spoilage event for pharma/frozen cargo	Configuration system already supports per-deployment threshold profiles; require sign-off on threshold sets per cargo category before go-live
No authentication on configuration changes in the current prototype	An unauthorized or accidental configuration change could silently widen thresholds or suppress alerts	Add signed/authenticated config updates restricted to a verified fleet-management backend; every change is already logged as a configuration_change event for audit
Power subsystem designed against datasheet specs, not measured against the real target vehicle fleet	An under-designed protection margin could let a transient damage the device, causing a monitoring outage	Bench-test the power subsystem against automotive transient standards (e.g. ISO 7637-2) and instrument a pilot fleet before wide rollout
Local event storage capacity can be exceeded during extended connectivity loss	Oldest events may be dropped by the retention policy before ever syncing, losing audit history for that period	Retention limit is already configurable; size SPI flash/SD capacity against worst-case offline duration and add a storage-nearly-full warning event
No firmware integrity protection (no secure boot / signing) in the current prototype	A compromised or corrupted firmware image could run undetected across a fleet, producing false readings or acting as an attack vector	Add secure boot and firmware signing before any production deployment — treated as a blocking item, not an optional improvement

7. Future Improvements

- Secure OTA firmware update architecture with staged rollout and rollback (Bonus 1).

- Predictive maintenance: trend battery voltage, sensor read latency, and vibration signatures over time to flag a likely refrigeration unit failure before a critical threshold is crossed (Bonus 6).
- A lightweight web/mobile dashboard surfacing live device status and historical events per container, aimed at non-technical operations managers (Bonus 5).
- A remote fleet-management architecture (device registry, bulk configuration push, health dashboard) designed explicitly for 50,000+ device scale (Bonus 4).
- A documented manufacturing test procedure for end-of-line validation before units ship (Bonus 3).
- Formal schematic capture and PCB layout once component selection is validated against real procurement/lead-time constraints.

End of report.